Software Engineering Class Task: Police Station Information Management System (PSIMS)

The Police Station Information Management System (PSIMS) was developed to replace traditional manual record-keeping practices in police stations with a computerized solution. Previously, FIRs, criminal records, evidence logs, and administrative reports were maintained using paper-based registers, which often resulted in delays, data inconsistencies, limited accessibility, and a high likelihood of human error. The primary objective of PSIMS was to automate these operations while improving efficiency, accuracy, and long-term operational effectiveness.

At the initial stage of development, the project team worked closely with police personnel to study existing workflows, legal documentation formats, and reporting requirements. These procedures were largely governed by established rules and regulations, allowing the development team to document system requirements in detail. However, as development progressed, additional requirements emerged due to changes in policies, updates in legal procedures, and practical feedback from system users. This indicated that system requirements needed to be reviewed and adjusted during the development lifecycle.

Rather than delivering the complete system at once, development was carried out in stages. Core functions such as FIR registration and basic record storage were implemented first, followed by more advanced capabilities including evidence tracking and administrative reporting. After each stage, feedback from users was reviewed and incorporated into subsequent development activities. This approach allowed the system to evolve gradually while remaining usable throughout the development period.

Managing potential risks was a continuous concern during the project. Since the system handled sensitive and legally significant information, issues such as data loss, incorrect record updates, and system failures were carefully considered. Design improvements and testing activities were carried out regularly to address these concerns before moving forward.

Testing and verification were performed throughout the development process. Individual components were examined before being combined with other parts of the system, and user feedback was used to confirm that system behavior matched real-world police operations. This helped identify and correct issues early, reducing the chance of serious problems after deployment.

Overall, this case study demonstrates how large-scale government information systems must be developed in a disciplined manner, where careful planning, controlled execution, and ongoing evaluation are essential to ensure that the final system remains dependable and sustainable over time.

Q- Based on the given case study, identify the Functional Requirements of the system. Also specify three Non-Functional Requirements: one Product Requirement, one Organizational Requirement, and one External Requirement.